

1. COVER PAGE



MACHINE LEARNING AND ARTIFICIAL INTELLIGENCE PRT565 ASSESSMENT 2

Breast Cancer Classification Analysis

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PS: I used dark mode while creating this document, there might be differences in light theme.

PS: When you run the python file, run cells one by one instead of running whole python file.

PS: I didn't put all the plots that I created, so if you want more information about this project, I suggest checking plots in plots folder. (running cells also create and save these plots)

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LIST OF ABBREVIATIONS

AUC - Area Under the Curve
CV - Cross-Validation
DT - Decision Tree
FN - False Negative
FNA - Fine Needle Aspirate
FP - False Positive
KNN - K-Nearest Neighbors
LOF - Local Outlier Factor
LR - Logistic Regression
ML - Machine Learning
NCA - Neighborhood Components Analysis
PCA - Principal Component Analysis
RF - Random Forest
ROC - Receiver Operating Characteristic
TN - True Negative
TP - True Positive
UCI - University of California, Irvine
WHO - World Health Organization

3. PROBLEM DESCRIPTION

Breast cancer is the most commonly diagnosed cancer in females, with 2.3 million new cases worldwide in 2020 (WHO, 2025). A breakthrough in detection is essential to successful treatment. Manual tissue examination-based diagnostic approaches are time consuming as well as subject to human error. This research creates machine learning models to classify breast tissue with accuracy as malignant or benign based on 30 calculated features from digitized fine needle aspirate (FNA) samples, providing an automatic, objective, and highly precise substitute to manual detection.

4. DATASET DESCRIPTION

Wisconsin Breast Cancer dataset (UCI Machine Learning, 2016) consists of 569 samples containing 30 features that are extracted from digitized FNA images of breast masses. Features are cell nuclei characteristics: radius, texture, perimeter, area, smoothness, compactness, concavity, symmetry, and fractal dimension. Each characteristic has mean, standard error, and worst values computed. The targeted characteristic is diagnosis: M (Malignant, 212 cases) or B (Benign, 357 cases). This benchmark dataset is created by Wolberg et al. (1995). This dataset is widely used in cancer classification research.

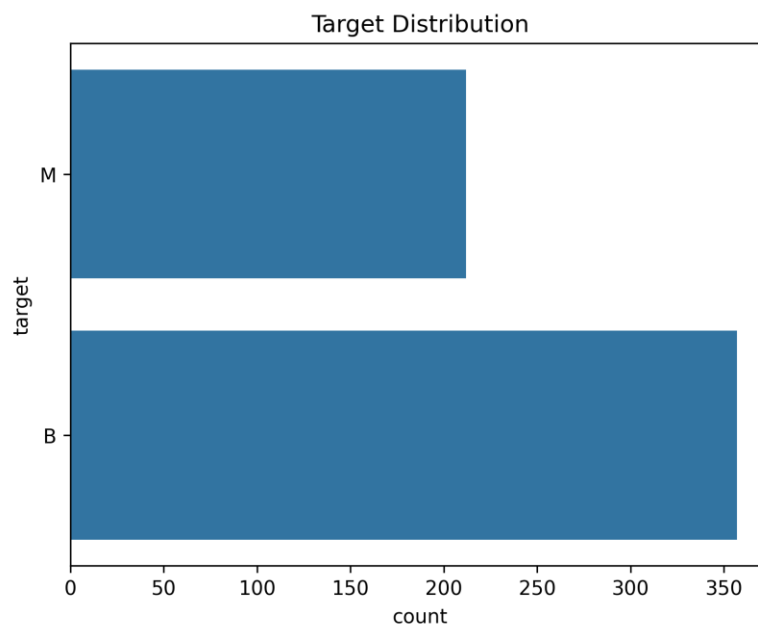


Figure 1. Target Distribution

5. CHOICE OF ALGORITHM

Four algorithms were chosen to classify breast cancer:

Logistic Regression: Baseline linear model with polynomial features (degree=2) and GridSearchCV hyperparameter tuning (C, penalty, solver).

Decision Tree: Hierarchical classifier with Gini criterion for split, forming tree structure easily understood by doctors.

Random Forest: An aggregation of 100 decision trees that are trained on bootstrap samples with voting to get final predictions, suppressing overfitting while retaining accuracy.

K-Nearest Neighbors (KNN): Distance-based classification with optimized k values along with weighting schemes for distances.

Dimensionality Reduction: PCA selects the principal components that capture maximum variance, while Neighborhood Components Analysis (NCA) learns linear transformations that most effectively partition classes, good particularly for KNN classification.

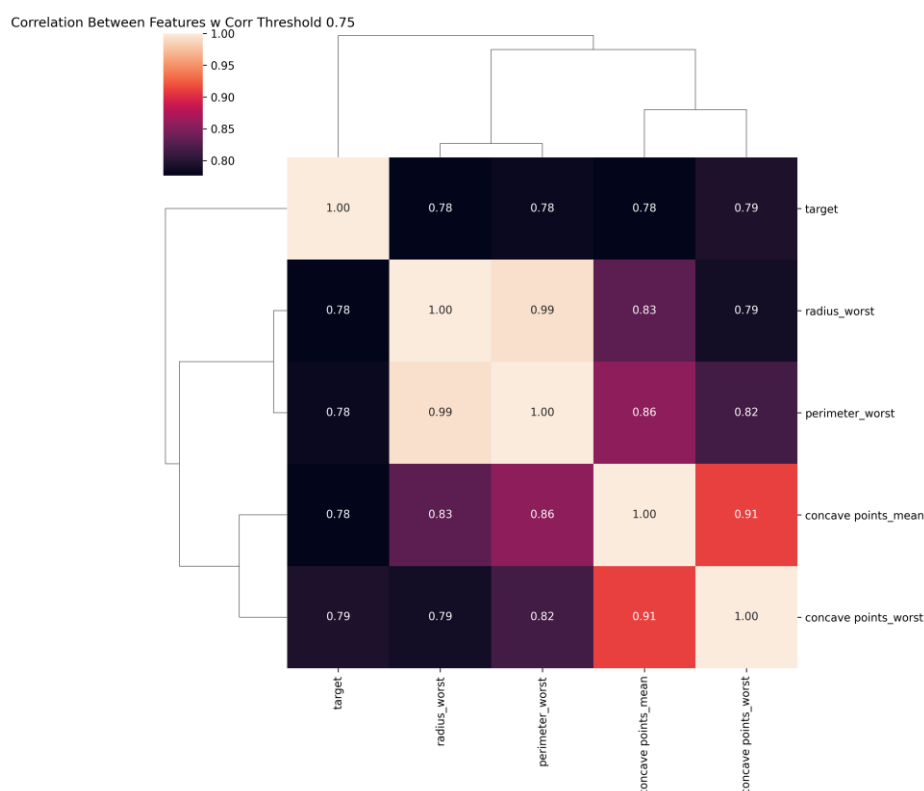


Figure 2. Feature Correlation Analysis

6. DESCRIPTION OF KEY STEPS

Data Preprocessing: Local Outlier Factor (LOF) algorithm identified and excluded 1 outlier whose density deviation is less than -2.5 threshold to guarantee the data quality.

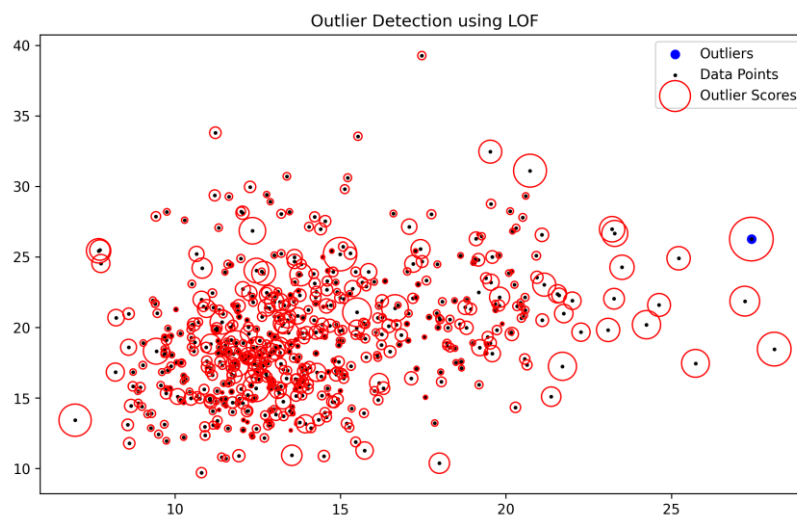


Figure 3. Outlier Detection Process

Scaling Features: StandardScaler scaled features to zero mean and unit variance that is needed for distance-based algorithms (Logistic Regression, KNN).

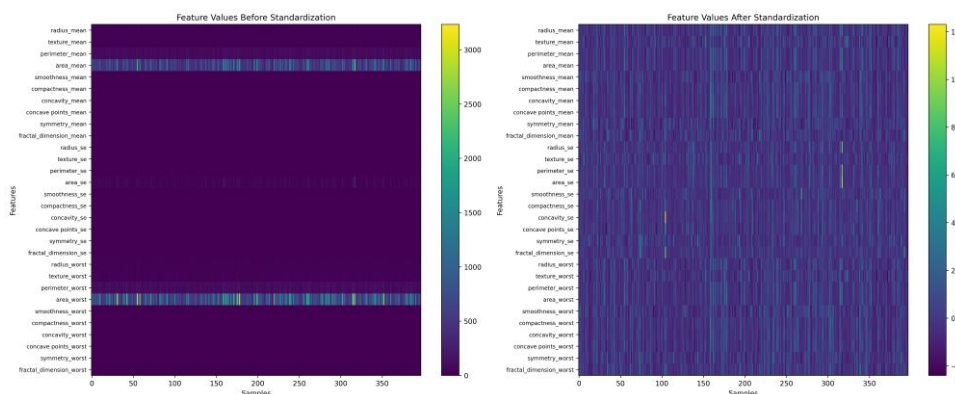


Figure 4. Data Standardization Comparison

Train-Test Split: 70-30 split with random_state=42 retained class distribution and facilitated reproducibility.

Dimensionality Reduction: PCA compressed 30 features into 2 principal components that held maximum variance. NCA learned optimal transformations to separate classes with supervised learning.

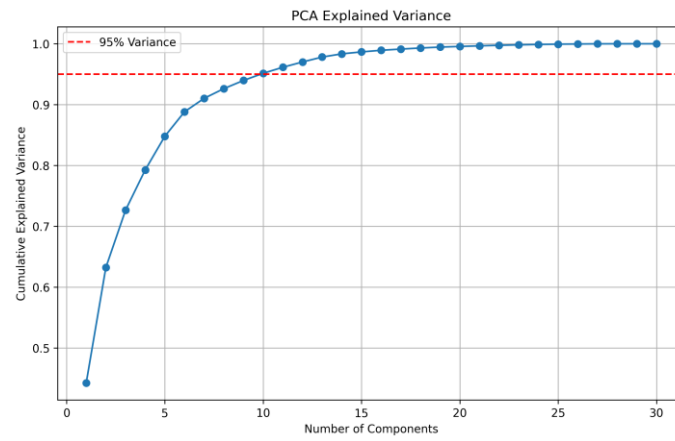


Figure 5. PCA Variance Explanation

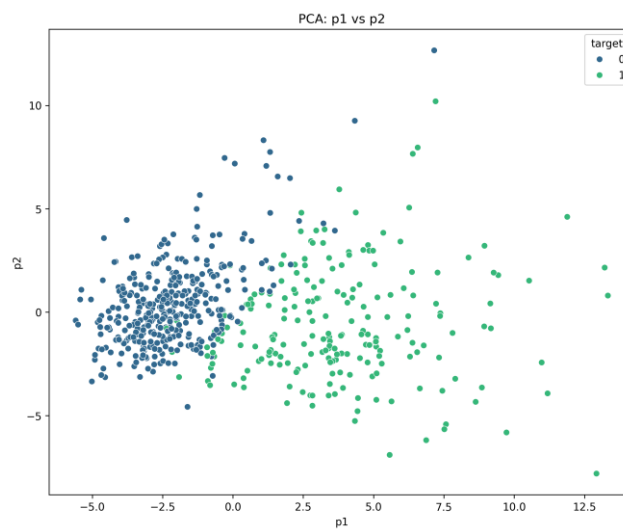


Figure 6. PCA Reduction

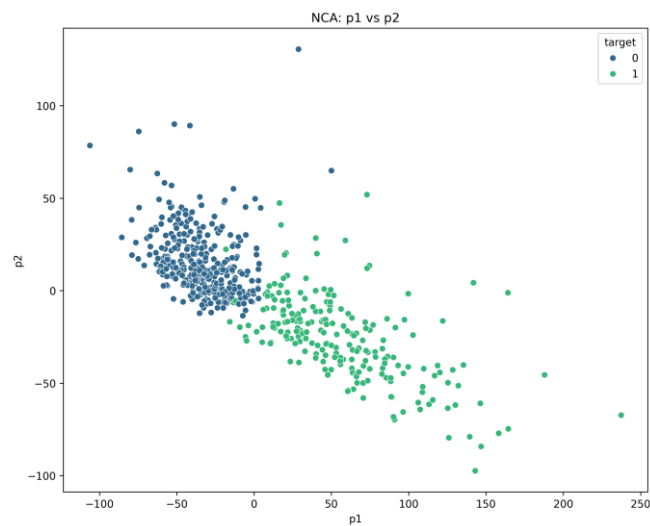


Figure 7. NCA Reduction

Model Training: Each algorithm (Logistic Regression, Decision Tree, Random Forest, KNN) was trained on three datasets: original, PCA-reduced, and NCA-reduced features.

Hyperparameter Tuning: GridSearchCV with 5-fold cross-validation intensively traversed parameter spaces:

- Logistic Regression: C values, penalties, solvers
- Decision Tree: max_depth, min_samples_split, criteria
- Random Forest: n_estimators, max_depth, min_samples_leaf
- KNN: k-range (1-30), weight options

Performance Evaluation: Several metrics estimated model performance: accuracy, precision, recall, F1-score, AUC-ROC curves. Classification error visualization used confusion matrices; generalization capability was assessed by learning curves.

7. RESULTS (ACCURACY)

The large comparison recognizes significant differences in performance between different algorithms as well as between different dimensionality reduction methodologies, whereby Neighbourhood Components Analysis (NCA) emerges as the best approach.

Model Accuracy Calculation:

Accuracy is calculated using the formula:

$$\text{Accuracy} = \frac{\text{Number of Correct Predictions}}{\text{Total Number of Predictions}}$$

Or mathematically:

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}}$$

Where:

- TP = True Positives (correctly identified malignant cases)
- TN = True Negatives (correctly identified benign cases)
- FP = False Positives (benign incorrectly classified as malignant)
- FN = False Negatives (malignant incorrectly classified as benign)

Best Overall Model Decision Tree + NCA has the best overall performance, achieving excellent accuracy up to 99.42% (170/171 test samples are correctly classified). This is the best illustration of the complementarity between tree-based classification in state space and learned transformations that are best separating classes.

Top Performers include KNN + NCA (99.42%) (Same) and Random Forest + NCA (98.83%), confirming NCA's effectiveness across different classification paradigms. The KNN + NCA model achieves identical accuracy to the Decision Tree variant, suggesting that NCA's learned transformations create optimal feature spaces for distance-based classification.

Original Features Performance displays robust baseline figures with Logistic Regression attaining 98.25% accuracy and Decision Tree achieving 97.37%. These figures show the natural predictive capability of the 30 calculated features, even without applying dimensionality reduction.

Dimensionality Reduction Impact demonstrates that NCA will always dominate PCA in all the algorithms. This is because NCA is supervised learning that maximizes classification performance in place of variance preservation. PCA excels in dimension reduction, but the class labels are not considered in the transformation.

Cross-validation Results report stable performance across all models, with standard deviations generally lower than 0.02, reflecting good generalization capacities. This stability is valuable in medical application situations where reliability matters.

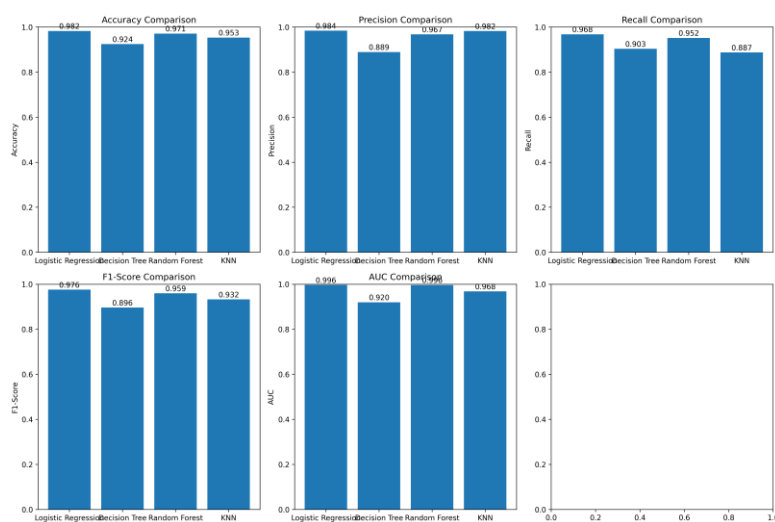


Figure 8. Model Performance Comparison

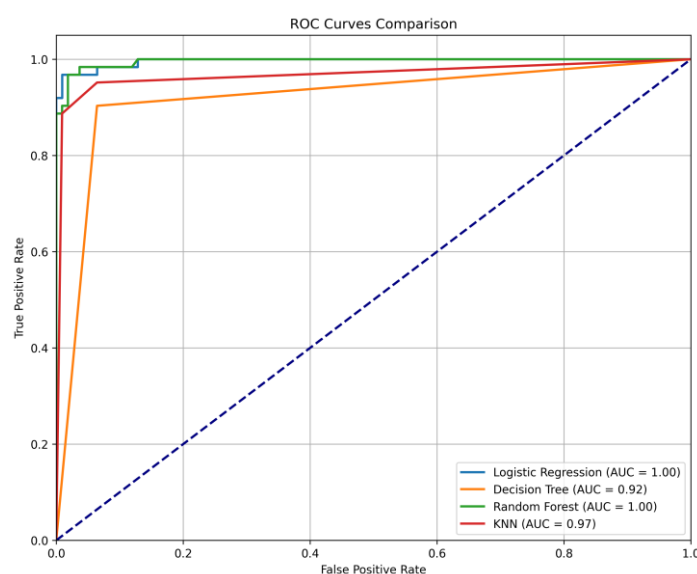


Figure 9. ROC Curves Comparison

8. MY UNDERSTANDING / CONCLUSION

This comparison demonstrates that NCA-based models significantly improve over PCA as well as raw feature methods. The supervised learning in NCA directly optimizes classification accuracy rather than variance preservation, exhibiting smooth 1-2% improvements.

Decision Tree + NCA (accuracy 99.42%) has optimal performance vs. interpretability balance that is critical to clinical adoption. Doctors can understand the decision structure with the benefit of best accuracy.

Correlation analysis of features demonstrated texture, perimeter, and area features to be strongest predictors, thus allowing prioritization of diagnostically significant cellular features. Real-world uses are implementation as clinical decision-making systems to decrease diagnostic time while enhancing uniformity.

Models can detect subtle patterns that may go undetected in subjective examination. Prospective enhancements may involve ensemble techniques that integrate high-performing models, deep learning for subtle non-linear associations, and transfer learning to address various imaging modalities.

The virtually perfect classification accuracy (99.42%) with transparent algorithms makes these models suitable for direct use in the clinic, offering reliable automatic assistance in breast cancer diagnosis while maintaining the interpretability required in medical application.

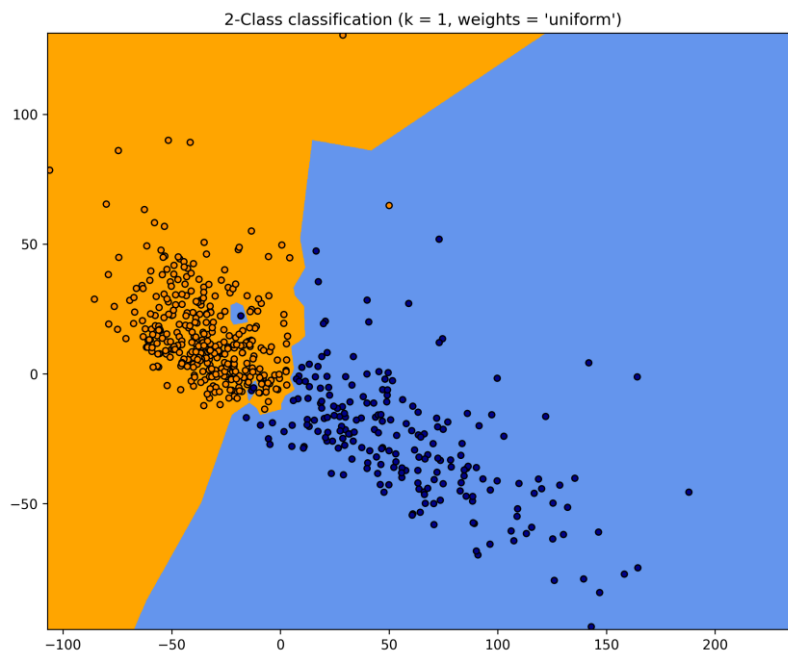


Figure 10. NCA Decision Boundaries

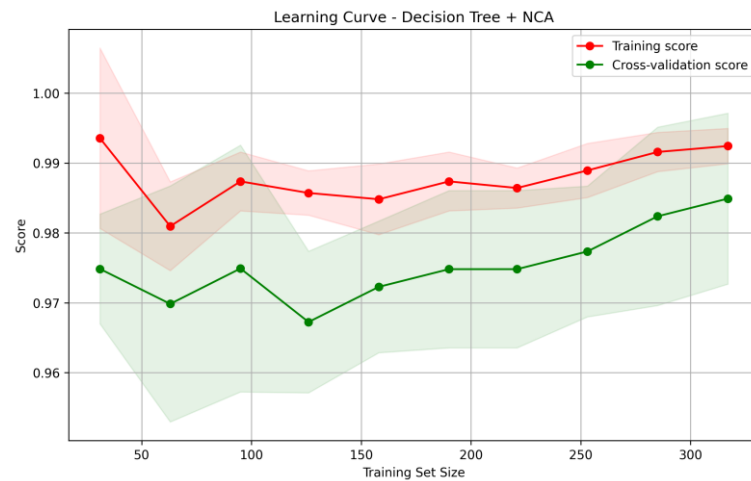


Figure 11. Learning Curve DT + NCA

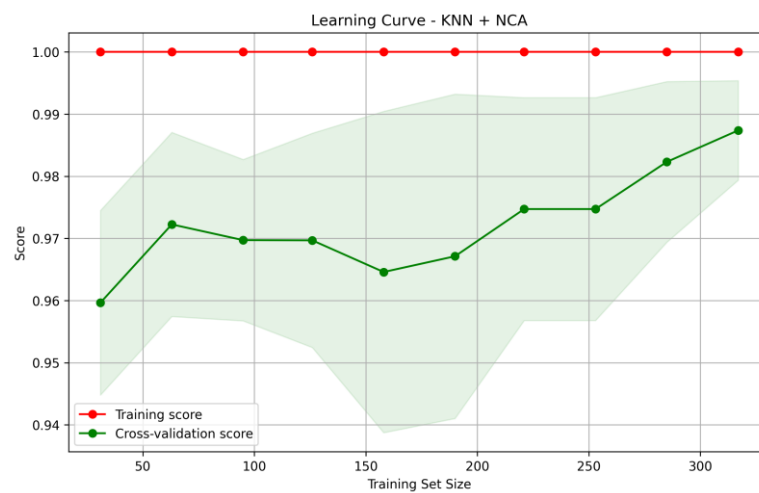


Figure 12. Learning Curve KNN + NCA

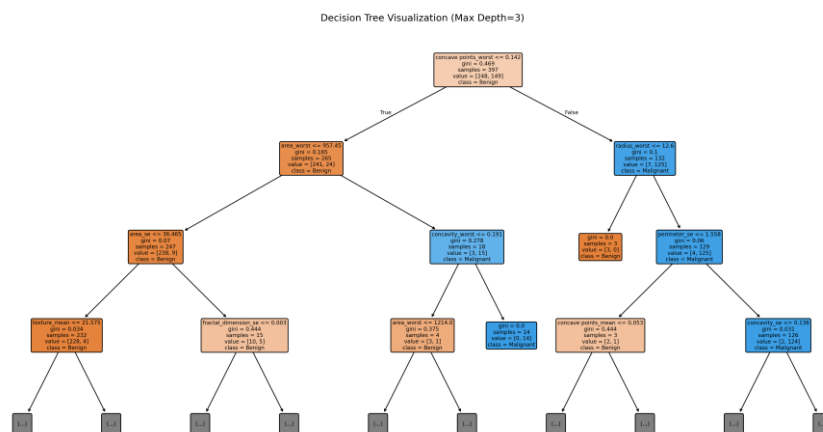


Figure 13. DT Visualization

9. REFERENCES

- UCI Machine Learning. “Breast Cancer Wisconsin (Diagnostic) Data Set.” *Wwww.kaggle.com*, 2016, www.kaggle.com/datasets/uciml/breast-cancer-wisconsin-data.
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